

Sarthak Upadhyay

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CAREER OBJECTIVE

Systems-oriented developer skilled in Python, Rust, and modern web technologies, with a focus on automation, CLI tooling, and open-source contribution. Actively seeking innovative projects and internship opportunities.

EDUCATION

Dayananda Sagar Academy of Technology and Management

Bachelor of Engineering in CSE, AI Specialisation

Bangalore, India

Sep. 2023 – Current

Fatima Sr. Sec. School

Higher and Senior Secondary Education

Gonda, India

2018 – 2022

EXPERIENCE

Green Intern

Sep. 2025 – Nov. 2025

1 Million 1 Billion (1M1B) Foundation

- 1M1B is a global non-profit empowering youth to build sustainable solutions; contributed as a Green Intern developing a VPS resource usage monitoring tool for cloud environments under their tech-for-good initiative.
- Built cross-platform system telemetry in Rust using Tokio for async data streaming, WinAPI and Linux /proc for native resource access, and JSON for standardized cloud-ready output, reducing monitoring overhead across hybrid VPS setups.
- Gained hands-on experience in cloud systems programming, async architectures, and responsible tech development aligned with environmental sustainability goals.

Open Source Mentor

Summer 2025

GirlScript Summer of Code (GSSoC) 2025

- Mentored contributors in an open-source program run by GirlScript Foundation, guiding participants through Python backend development, lexer implementation, and real-world codebase workflows.
- Assisted with team management, reviewed pull requests, provided technical feedback, and helped onboard new developers to open-source collaboration during the summer cohort.

Member, Android Application Developer

June 2024 – Present

Maker Space Club

DSATM, Bangalore, India

- Built an Android application managing calls and battery health optimisation using React Native and Android SDK APIs for native device integration.
- Improved and corrected existing software and system applications, performing bug fixes in routing and API interactions.
- Conducted studies on nuanced features of unique platforms to lead team in development of applications tailored to Android and VMs.

PROJECTS

Surge

2025

Stack Used: SvelteKit, Svelte 5, TypeScript, TailwindCSS, Supabase, Python, Leaflet

- Built a real-time environmental monitoring platform tracking AQI and WQI across 20 Bangalore localities with interactive Leaflet heatmaps, live SSE data streaming, and historical trend charts.
- Implemented a Prophet-inspired time-series forecasting algorithm for 24h and 48h AQI and WQI predictions with uncertainty bands, confidence scoring, and seasonal factor modeling.
- Designed a Supabase PostgreSQL backend with JSONB schemas, indexed queries, and SvelteKit serverless API routes for real-time data ingestion, historical lookup, and prediction endpoints.

Wisp

Nov 2024 – Present

Stack Used: Rust, Tauri, Svelte, Bootstrap Icons

- Built a lightweight, cross-platform text editor using Rust and Tauri for native performance and enhanced desktop application security.
- Designed a clean, responsive UI with Svelte and TailwindCSS for a fast, distraction-free editing experience.
- Integrated CodeMirror for syntax highlighting, customizable keybindings, and smooth multi-language editing support.

LLM Bias Benchmark

Jun 2024 – Present

Stack Used: Python, JSONL, Pandas, Matplotlib

- Built an end-to-end benchmark pipeline to evaluate cultural and linguistic bias in LLMs across 5 languages (Hindi, Arabic, Spanish, Chinese, Russian).
- Designed a unified model interface with a template-based data pipeline for prompt expansion, localization, and human annotation tooling.
- Implemented mitigation strategies including prompt engineering, fine-tuning, and post-processing; visualized results via heatmaps, charts, and case studies.

Synap

Sep 2025 – Nov 2025

Stack Used: Rust, Tokio, WinAPI, Linux /proc

- Built a cross-platform system information fetcher in Rust with native support for Linux and Windows, enabling unified monitoring across hybrid infrastructures.
- Implemented async data streaming via Tokio runtime with direct integration to Windows WinAPI and Linux /proc, outputting standardized JSON telemetry at configurable intervals.
- Reduced monitoring overhead by replacing multiple platform-specific tools with a single memory-safe binary leveraging Rust’s zero-cost abstractions.

DeepSeq

2025

Stack Used: Python, Streamlit, ESMFold, Py3Dmol, scikit-learn

- Developed an AI-powered protein analysis tool offering 3D structure prediction, sequence embeddings, residue property analysis, stability assessment, and function prediction.
- Integrated ESMFold API for high-accuracy structure prediction and ESM2 transformer models to generate high-dimensional protein sequence embeddings with PCA/t-SNE visualization.
- Built an interactive Streamlit interface with Py3Dmol for real-time 3D molecular visualization supporting multiple rendering styles, color schemes, and downloadable results.

TECHNICAL SKILLS

Languages: Python, Rust, TypeScript, JavaScript, C, Java, HTML, CSS, SQL

Frameworks: Svelte (Kit), FastAPI, Rocket.rs, Flask, Express.js, React, Next.js, Django, React Native

Databases: PostgreSQL, MySQL, MongoDB, SQLite3, Redis, Supabase

Developer Tools: Git, GitHub, VSCode, Zed, Neovim, Bash/Zsh/Fish shells, Vercel, Render

Libraries: pandas, NumPy, Matplotlib, Tensorflow, Keras, asyncio

CERTIFICATIONS

The Complete Full-Stack Web Development Bootcamp (by Angela Yu) — Udemy

Jul 2025

Everything You’ll Need to Know About Git (by The Primeagen) — Frontend Masters

Mar 2025

Prompt Engineering Safety & AI Engineering Safety Expert Certification & Assessment — MTF Institute

Jul 2025

Hands on GitHub Action Event (by Microsoft Student Ambassadors program)— Github Education

Oct 2023